

Don't Trash It - Let AI Sort It with 89% accuracy

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DS 4002:
April 24th, 2026



Project Details



The problem: “Single stream” recycling programs (where all types of materials are collected in one bin) often fail because they prioritize ease of use and collection volume over quality, leading to contamination issues.

Creating a computer vision tool will allow us to handle these co-mingled materials, helping sorting facilities recover more resources despite high contamination levels.

Project Details



Hypothesis: During image classification, we predict that ResNet-50 will perform the best out of all of the models in terms of speed and accuracy.

Research Question: Can an image classification model accurately predict the correct disposal category for waste based on visual features?

Modeling Approach: Three different pre-trained models, MobileNetV3-Small, EfficientNet-B0, and ResNet-50, each fine-tuned on the dataset using transfer learning. The backbone weights are frozen first to train the classification head. Afterwards, the top layers are unfrozen, and more training occurs, but at a lower learning rate. Model performance will be evaluated using a confusion matrix, F1 scores, and by analyzing the speed of the classifications

Goal statement: Compare three neural network models to classify landfill waste into nine categories, with a target of at least 50% accuracy, in order to improve the efficiency of automated recycling systems.

Data Acquisition

- Real Waste Dataset obtained from the UC Irvine Machine Learning Repository
 - Creative Commons License - free to be shared or adapted for any purpose
- 4752 images of types of waste captured at point of reception in a landfill environment
- Categorized into nine different material types: Cardboard, Food Organics, Glass, Metal, Miscellaneous Trash, Paper, Plastic, Textile Trash, and Vegetation

Cardboard



Food Organics



Glass



Metal



Miscellaneous Trash



Paper



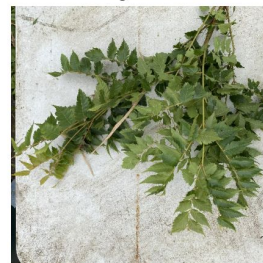
Plastic



Textile Trash



Vegetation



Analysis Plan and Justification



Data Preparation

Images are resized, normalized, and split into training, validation, and test sets (70/15/15).



Data Augmentation and Class Balancing

Data augmentation simulates real-world variation in lighting and orientation, while weighted cross-entropy loss addresses class imbalance.



Model Training (Transfer Learning)

MobileNetV3-Small, EfficientNet-B0, and ResNet-50 are fine-tuned via two-phase transfer learning: first training only the classification head, then unfreezing the top layers at a reduced learning rate



Model Evaluation

Each model is evaluated on the test set using macro-averaged F1 score, a confusion matrix, and inference speed per image.

Bias and Uncertainty Justification

Class Imbalance

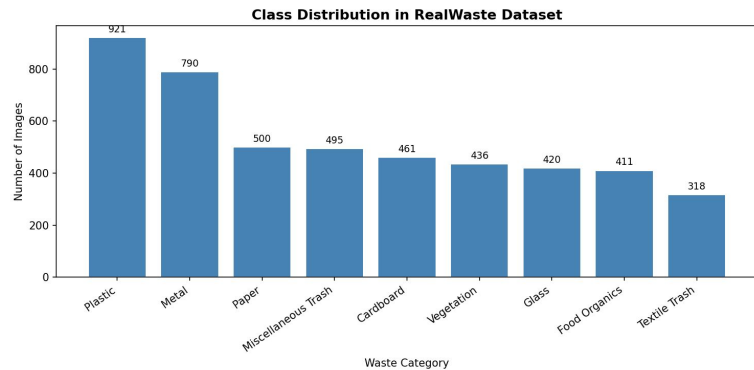
- The distribution of the nine waste categories is highly imbalanced.
- Applied a weighted cross-entropy loss to prevent model from defaulting to most common class

Isolated Subject Matter

- Images feature isolated items on concrete, unlike the co-mingled reality of single-stream recycling.
- Expanded training data using flips, rotations, and color changes to simulate diverse real-world conditions

Limited Dataset Size

- The dataset contains a relatively small pool of 4,752 total images to train deep learning models across nine distinct classes



Tricky Analysis Decision

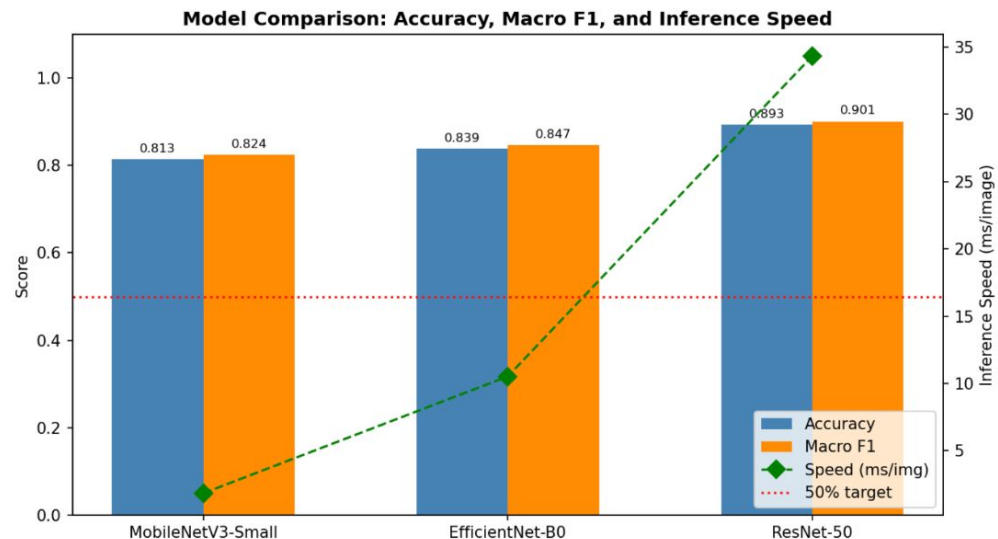


Elevating Speed as a Primary Metric

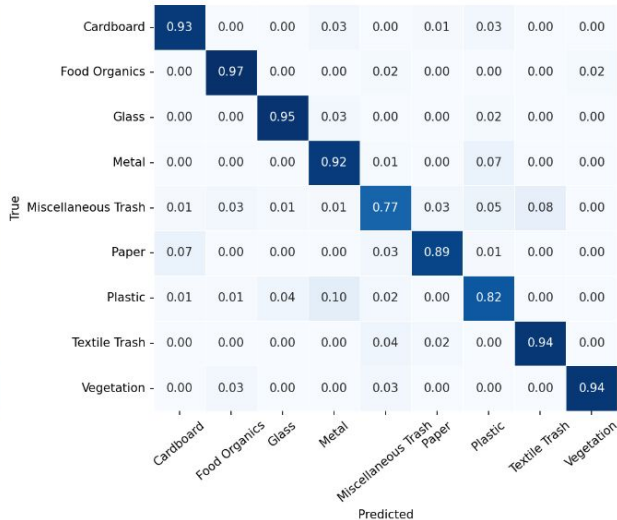
- **The Standard:** Most image classification projects solely chase the highest possible accuracy.
- **The Tricky Part:** In a physical, automated recycling facility, a model that is highly accurate but slow creates a massive physical bottleneck.
- **The Decision:** We determined that pure accuracy isn't the only metric that matters. We chose to elevate processing speed to a primary metric alongside our F1 scores, prioritizing real-world application over lab-environment perfection.

Results and Conclusions

- All three models beat our 50% goal, with ResNet-50 performing best at 89% accuracy, confirming our hypothesis
- MobileNetV3-Small is the speed winner at ~19x faster than ResNet-50, so there's a real accuracy vs. speed trade off depending on the use case
- Vegetation was easiest to classify, Miscellaneous Trash was hardest



MobileNetV3-Small
Macro F1=0.824 Acc=0.813



Next Steps



Test on additional datasets

- Test how well the model generalizes beyond Real Waste
 - Collect data from different landfill facilities

Improve Miscellaneous Classification

- Relabel ambiguous items, to remove this category completely
- Focus on better labeling for hard to classify edge cases

Evaluate on Real-Time Video Feed

- Test the model on live conveyor belt footage rather than static images to simulate actual deployment conditions

References



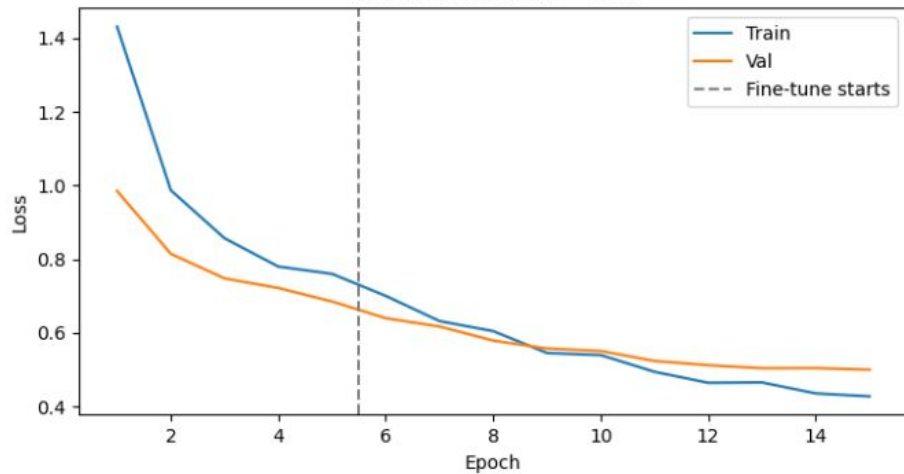
- [1] U.S. Environmental Protection Agency, "Recycling Basics and Benefits," EP A, [Online]. Available: <https://www.epa.gov/recycle/recycling-basics-and-benefits>. [Accessed: Apr. 3, 2026].
- [2] N. Hutman, "Single-Stream Recycling Doesn't Work, Institute for Local Self-Reliance, [Online]. Available: <https://ilsr.org/articles/single-stream-recycling-doesnt-work/>. [Accessed: Apr. 3, 2026].
- [3] S. Single, S. Iranmanesh, and R. Raad, "RealWaste: A Novel Real-Life Data Set for Landfill Waste Classification Using Deep Learning," Information, vol. 14, no. 12, p. 633, 2023. [Online]. Available: <https://archive.ics.uci.edu/dataset/908/realwaste>.

Github: <https://github.com/patrickzyin/DS-4002-Project-3>

Questions?



MobileNetV3-Small — Loss



MobileNetV3-Small — Accuracy

